BAR-ILAN UNIVERSITY

Dental Pathology Detection Using Deep Learning

Aviel Eliyahu

Submitted in partial fulfillment of the requirements for
the Master's Degree in the Faculty of Engineering,
Bar Ilan University

Ramat-Gan, Israel 2026

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This work was carried out under the supervision of Prof. Aryeh Weiss, Faculty of Engineering, Bar-Ilan University.

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Abbreviation and acronyms

1. IDF – Israel Defense Force
2. CNN – Convolutional Neural Networks
3. GPU - Graphics Processing Unit
4. JSON – JavaScript Object Notation
5. ML – Machine Learning
6. AI – Artificial Intelligence
7. RO – Radiopaque
8. RL - Radiolucent

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Abstract

This study is a retrospective study whose goal is to train a deep learning model capable of detecting dental pathologies and findings in dental X-ray images. In the study, six dentists annotated more than 5,500 bitewing and periapical images to train the deep learning model. The chosen model is a U-Net, which utilized pre-trained weights (transfer learning) and employed callbacks to stop training before overfitting occurred, as well as to save the model and perform other tasks. The evaluation results show that the model detected high contrast features well, but performance on low contrast/diffuse classes (such as carries and bone loss), while detected, were not detected accurately enough for clinical use at this point. Implementing the model could pave the way for annotating additional images and further improving the model.

1. Introduction

Teeth are one of humans' most noticeable features. Our teeth are made of enamel – the hardest substance in the human body [1]. Like all human organs, teeth also suffer from various diseases, such as caries (tooth decay), gingival recession, and more.

Dental pathologies are usually diagnosed via X-ray imaging, using what is known as a bitewing radiographic technique [2]. In this procedure, the patient is given a film-holder with a wing to bite. At the end of the bite tab there is a film/sensor which can detect X-ray radiation and produces the image. This type of imaging allows dentists to assess both the mandibular (lower jaw) and maxillary (upper jaw) teeth.
Another popular diagnostic tool is the intraoral periapical radiograph, which focuses on either the mandibular or maxillary teeth, and shows the root of the teeth.

Dentists sometimes need to check tens if not hundreds of patients in a single day, carefully examining the different bitewing and periapical images, looking for signs and indications of diseases. This daunting task – which requires careful observation and expertise – can be quite overwhelming and lead to wrong diagnosis, and sometimes wrong treatment [3].

Image processing in general – and deep learning in particular – has shown great promise in detecting anomalies in medical imaging, as well as dental imaging [4].

The goal of this research was to develop a deep learning image segmentation model for detection of pathologies and other relevant findings in dental radiography. Our model is designed to be a decision support model, one which will help dentists to detect dental pathologies faster, and with fewer errors.

2. Background

2.1. Objective

The goal of this thesis is to develop a deep learning model which can identify and segment pathologies in dental x-rays images. Since almost every patient that visits a dentist undergoes bitewing imaging, dentists may be overwhelmed by the number of radiographs they need to examine. The large number of images, inexperience, lack of sleep, extended working hours and stress may lead to misclassification and errors. This is especially true for mass screening scenarios, such as within the army, where every recruit undergoes dental imaging, and dental records are stored for each recruit.

With a decision support model, the dentist could run the model on the dental x-ray images and spend more time on those images that are flagged by the system as having possible pathologies. This can be expected to improve the accuracy of the large-scale screening, as is done by the army.

2.2. Dental Anatomy

This subsection is based on refence [1]. Human teeth are one of the hardest organs in the human body. The tooth consists of four tissues as shown in Figure 2-1: Enamel, Dentin, Cementum, and Pulp.

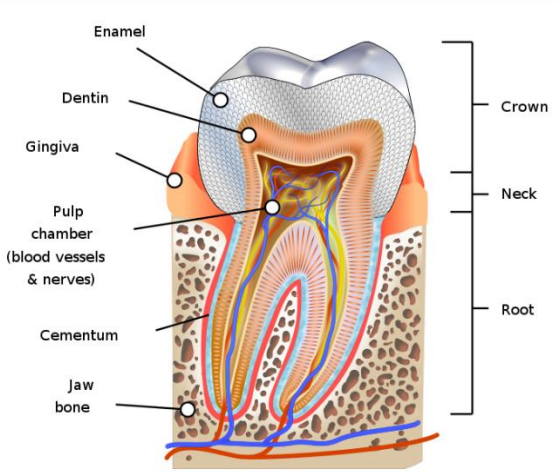


Figure 2-1 Tooth anatomy

Adapted from: "Human tooth diagram-en.svg" from Wikimedia Commons by K. D. Schroeder, CC-BY-SA 4.0

https://upload.wikimedia.org/wikipedia/commons/4/43/Human_tooth_diagram-en.svg

Of the four, only two of these tissues are visible in a normal intact tooth: enamel and cementum. The other two tissues (dentin and pulp) are usually not visible on a healthy intact tooth.

Enamel, dentin, and cementum are relatively hard due to their considerable mineral content, especially calcium.

Enamel is the white protective layer of the tooth. It is the hardest substance in the human body, consisting of 95% calcium hydroxyapatite which gives it its hardness.

Cementum is seen as a very thin external layer of the tooth root. The cementum is comprised of about 65% calcium hydroxyapatite, 35% organic matter (collagen fibers), and 12% water. The cementum is softer than the enamel but as hard as bones.

Dentin is the underlying surface of the enamel and cementum and is the largest tissue of the inner portion of each tooth crown and root. It originates from the pulp cavity in the center of the tooth and extends to the inner surface of the enamel (on the crown) or cementum (on the root). Normally dentin is not visible, unless the enamel or cementum has been damaged. Mature dentin is composed of about 70% calcium hydroxyapatite, 18% organic matter (collagen fibers), and 12% water, making it harder than cementum but softer and less brittle than enamel.

Pulp is a soft tissue that contains nerves and blood vessels, Like the dentin, the pulp is not normally visible in a healthy tooth. It has several functions such as storing dentin producing cells, transmitting signals via nerves, supplying nutrients and oxygen via blood vessels, and responding to injury or decay by producing dentin.

2.3. Dental Pathologies

Teeth are susceptible to damage and disease, many of which can be observed using bitewing imaging. Pathologies that are diagnosed using bitewing images include the following:

2.3.1. Dental Caries

Dental caries – or tooth decay – is a dental lesion, and one of the most prevalent chronic diseases worldwide. Caries form through a complex interaction over time between acid-producing bacteria and fermentable carbohydrates, and many host factors including teeth and saliva [5]. In addition to damaging the teeth themselves, tooth decay can also hurt self-confidence of the person suffering from it.

As caries advances, it can lead to the formation of cavities in the teeth. Dental caries progression can be halted at almost any stage, even when the dentin or enamel has been significantly compromised, provided that sufficient dental plaque is effectively removed. This underscores the importance of regular dental check-ups, proper oral hygiene practices, and dietary choices to prevent and manage dental caries, thereby preserving oral health and well-being.

An example of dental caries is shown in Figure 2-2.



Figure 2-2 - Dental caries

Adapted from https://commons.wikimedia.org/wiki/File:Dental_Caries_Cavity_2.JPG

2.3.2. Periodontitis

Periodontitis is a serious gum infection that damages the soft tissue and, if untreated, can destroy the bone supporting the teeth, as can be seen in Figure 2-3. It is caused by bacteria that accumulate on the teeth and gums, leading to inflammation and the formation of pockets

between the teeth and gums. Over time, these pockets can deepen, causing tooth loss and other health complications. Periodontitis affects all of the periodontal tissues. Periodontitis is preventable with good oral hygiene and treatable through professional cleaning and, in advanced cases, surgical interventions. Early detection and treatment are crucial to prevent permanent damage [6].

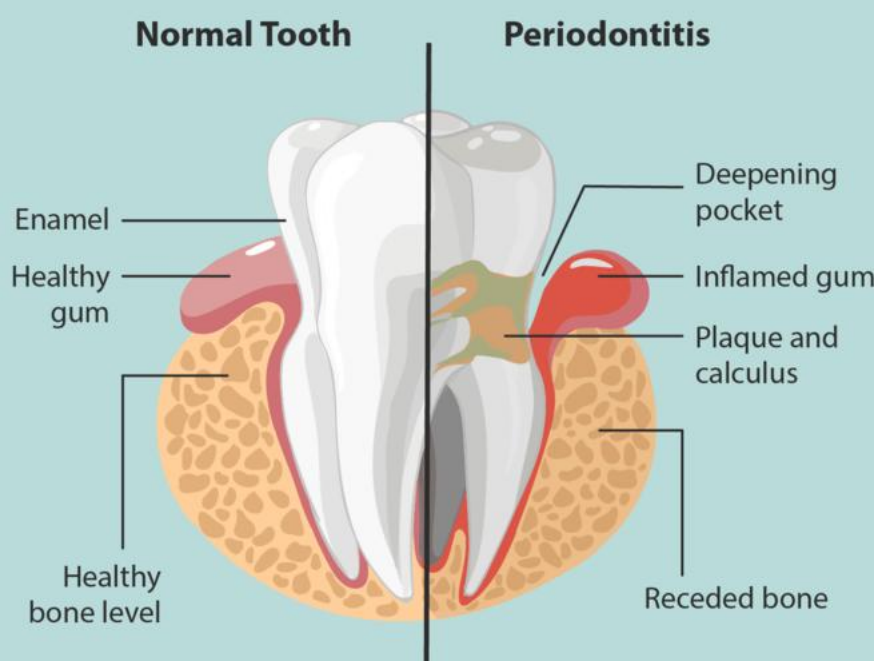


Figure 2-3 - Periodontitis

Adapted from https://commons.wikimedia.org/wiki/File:Depiction_of_a_Periodontitis_patient.png

2.3.3. Impacted Teeth

Impacted teeth are defined as teeth that fail to erupt into the dental arch within the expected developmental timeframe due to obstruction by adjacent teeth, dense overlying bone, excessive soft tissue, or inadequate arch length [7]. Tooth impaction is a common clinical condition, studies report prevalence rates ranging from 2.9% to 13.7% depending on ethnicity and study populations. Third molars account for most of impacted teeth, with mandibular third molars more frequently affected than maxillary counterparts. Excluding third molars, maxillary canines are the most frequently impacted teeth, followed by mandibular canines.

The most common complications of impacted teeth include odontogenic infections (pericoronitis, caries), follicle enlargement, cyst formation, resorption of adjacent tooth roots, and alveolar bone loss.

2.4. Dental Treatment

2.4.1. Restoration

Dental restoration (also known as dental fill) involves the care for oral and dental tissue. It is the process of repairing or replacing damaged, decayed, or missing teeth to restore their normal function, structure, and appearance [8] as seen in Figure 2-4. This can involve various procedures, including fillings, crowns, bridges, implants, and dentures, depending on the extent of the damage and the specific needs of the patient. Modern dental restoration techniques also focus on using materials that closely mimic the natural look and feel of teeth (such as Amalgam, Adhesive restoration, Resin composite and more), ensuring both functionality and aesthetics are maintained.



Figure 2-4 Dental Filling
Adapted from https://www.bleudentist.com/dental-filling-facts-in-2019/

2.4.2. Implants

A dental implant involves embedding an artificial root, typically made of titanium alloy, into the jawbone to replace missing teeth, as seen in Figure 2-5. This provides a durable and natural-looking alternative to dentures or bridges. After three to six months of healing following the surgical placement, the implant can be used to anchor a crown or a screw-retained fixed dental prosthesis, or to support a removable partial denture. As the implant gradually fuses with the bone, it forms a strong and stable foundation for these attachments. Multiple implants can also be used to replace several missing teeth, either with a fixed dental prosthesis or a removable partial denture [1].



Figure 2-5 - Dental implant

Adapted from: https://commons.wikimedia.org/wiki/File:Dental-implant-illustration.jpg

2.4.3. Crowns

Dental crowns are tooth shaped protective caps placed over damaged or weakened teeth to restore their shape, size, strength, and appearance. They are commonly used to cover a tooth that has undergone significant decay, has a large filling, or has undergone a root canal procedure [1]. Crowns can also be used to cover a dental implant or to improve the appearance of a discolored or misshapen tooth. Made from materials like porcelain, metal, or a combination of both, dental crowns are designed to blend in with natural teeth, providing both functional and aesthetic benefits. A typical crown is shown in Figure 2-6.



Figure 2-6 - Dental Crown

Picture by Prisach wikiinfounder the Creative Commons Attribution-Share Alike 4.0 International license, adapted from: https://upload.wikimedia.org/wikipedia/commons/d/d9/Dental-crown.jpg

2.5. Intraoral Imaging and Pathology Detection

2.5.1. X-Ray Basics

This section is based on reference [6].

X-Ray imaging is based on the interaction of high energy photons with matter.

Photons are composed of an electric field and a magnetic field both propagating via a medium, while the electric field and the magnetic field are perpendicular to one another. These photons can be described using sine waves and characterized by their wavelength and frequency. X-Rays are electro-magnetic waves with a wavelength of less than 1 nano meter ($10^{-9}[m]$).

All the matter in the universe is made of atoms, atoms have a nucleus composed of protons and neutrons, and an outer shell of electrons.

When X-Rays pass through a patient (or, through his atoms) there could be three different interactions: absorption, reflection, and transmission. Usually, soft tissues (like cheeks and gum tissue) transmit most of the X-Rays, while denser tissues (like bones and teeth) tend to absorb the radiation. Some absorption interactions use all of the incoming photon's energy, while some interactions use a fraction of the energy, causing the photon to scatter. This difference in absorption leads to differences in attenuation, which in turn is the reason for the varying intensities seen in an X-Ray image (for example, more absorbent tissues result in lighter areas in the x-ray image).

2.5.2. Intraoral Imaging

Intraoral imaging refers to dental radiographic techniques conducted within the oral cavity to visualize tooth structures and surrounding tissues. One of its primary applications involves the detection of interproximal caries, which are lesions that develop in the contact areas between adjacent teeth—commonly referred to as the interproximal spaces [1].

Intraoral imaging is widely used to detect interproximal caries and evaluate the height of alveolar bone as seen in Figure 2-7.

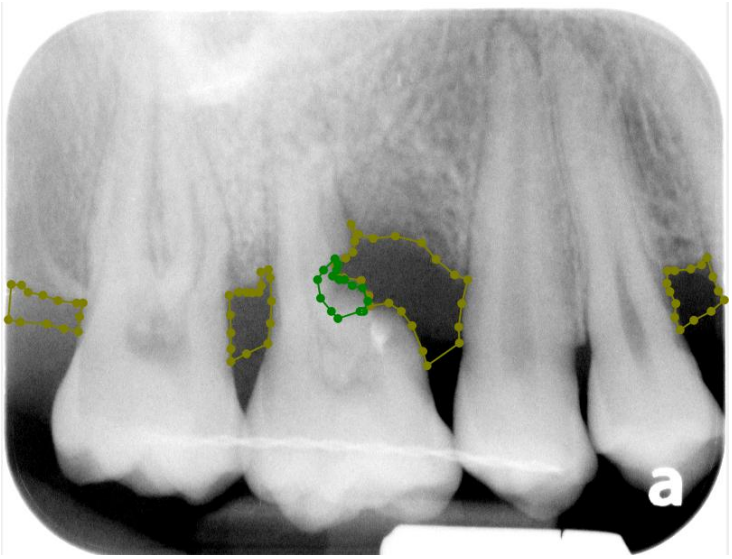


Figure 2-7 - Periapical image

Bitewing images capture both the maxillary (upper jaw) and mandibular (lower jaw) teeth in one image, as opposed to Periapical imaging which shows only the maxillary or mandibular teeth.

Intraoral imaging is especially important for early detection of interproximal caries, even before they become clinically visible to visual inspection. It can also reveal secondary caries below restorations (tooth filling) that may not be detected in the periapical views.

Intraoral imaging projections are also useful for evaluating the periodontal (near teeth) condition. Changes in bone height can be assessed accurately by comparison with the adjacent teeth as is shown on Figure 2-8, where the red line represents the true height of the bone, while the blue line represents the bone's natural height.

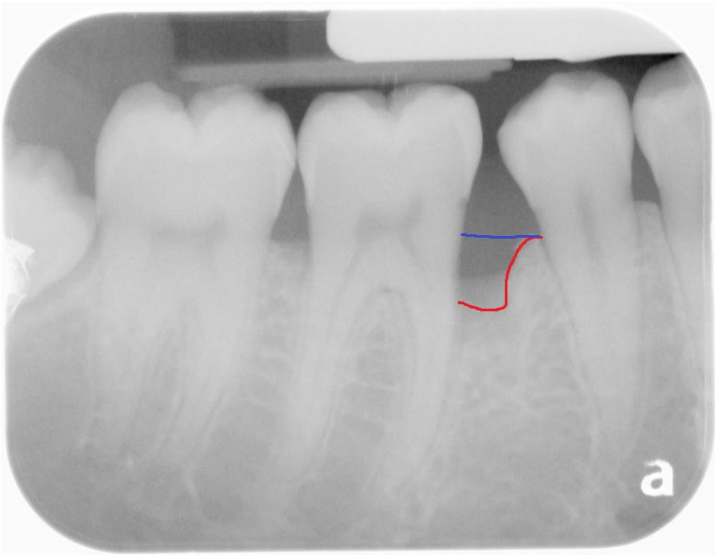


Figure 2-8 - Periapical image with bone loss

2.5.3. Interpretation

One crucial skill that any dentist acquires is interpreting dental X-ray images. There are two main components for this task: perception, the ability to recognize abnormal patterns in the image, and cognition, the interpretation of these abnormal patterns to arrive at a diagnosis [6].

2.5.3.1. Caries

Radiography (and specifically, bitewing imaging) is a very popular method for detecting dental caries (which were reviewed in 2.3.1). Caries causes demineralization of enamel and dentin, which therefore absorb fewer x-ray photons as compared to healthy enamel and dentin. Therefore, they are seen in a diagnostic image as a radiolucent (darker) zone [6].

Radiography is especially important when the tooth itself is still intact, and the caries are not visible to the naked eye.

The shape of the early caries in the enamel is often a triangle with its broad base at the tooth surface spreading along the enamel, but other appearances are also possible, as seen in Figure 2-9.

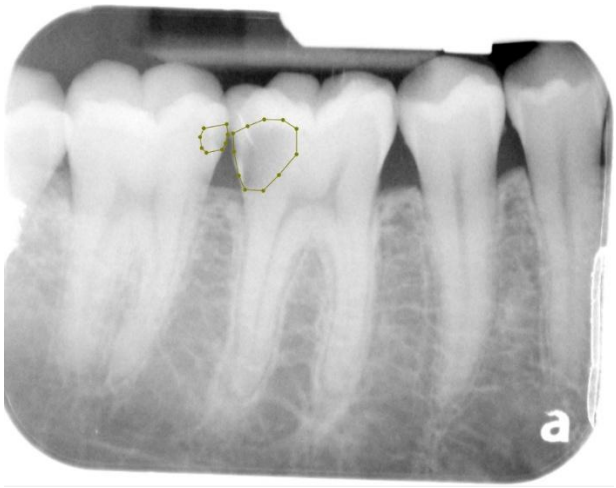


Figure 2-9 - Dental caries

Even experienced dentists often do not always agree on the presence or absence of carious lesions when examining the same set of images [9]. The disagreement could be due to the morphology of the tooth that may mimic the appearance of a carious lesion, or due to overlapping contact points in that may obscure a lesion. These factors are compounded by fact that caries are usually low contrast objects that may be missed upon cursory examination.

A study found that by using the conventional clinical and radiographic methods, 50% of caries will not be diagnosed by dentists [5].

Secondary lesions are carious lesions developing near an existing restoration. This type of caries may be obscured by the radiopaque image of the restoration as shown in Figure 2-10. Again, identification is hindered by the diffuse, low contrast nature of this pathology.

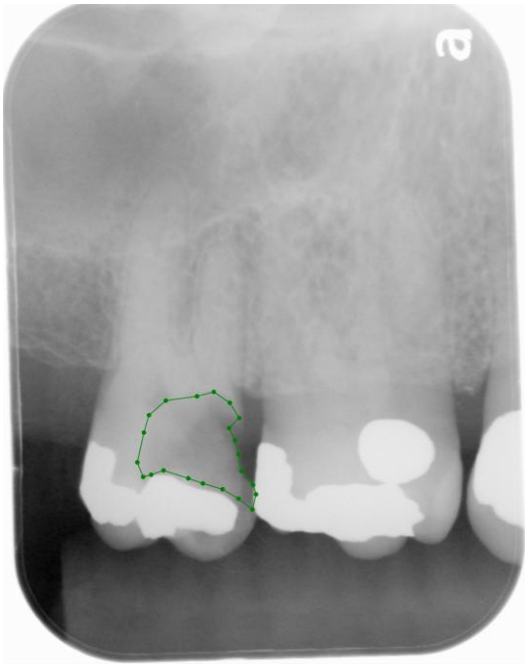


Figure 2-10 - Secondary caries

2.5.3.2. Periodontitis

Intraoral images may provide an incomplete presentation of the status of the periodontium and periodontitis, which were reviewed in 2.3.2.

They have the following limitations:

1. Since this imaging technique produces a 2D image from a 3D structure, some key elements may overlap and could be obscured, and only the interproximal bone is seen clearly. However, changes in the density of the root structure, which appears more radiolucent, could help dentists diagnose bone loss.
2. Early detection is difficult, since the earliest destructive lesions in bone do not cause a sufficient change in density to be detectable.

The normal alveolar bone supporting the teeth has a distinctive appearance, typically covered by a thin layer of opaque cortical bone at the alveolar crest as can be seen in Figure 2-1. The height of the crest is usually about 0.5 to 2.0 mm below the cementoenamel junctions (CEJs) of adjacent teeth. In the posterior region, the alveolar crest is parallel to a line connecting the CEJs of neighboring teeth. In the anterior region, the crest is often pointed and may have a clearly defined cortical layer. A well-mineralized cortical outline of the alveolar crest generally suggests the absence of active periodontitis. However, a less mineralized crest can be present in individuals with or without periodontitis.

In periodontal disease, diagnostic images reveal changes in the bone that are like those seen in other inflammatory bone diseases. These changes can be categorized into alterations in the morphology of the supporting alveolar bone and modifications to the bone's internal density and trabecular pattern. Morphological changes often result from the loss of interproximal crestal bone or bone covering the buccal or lingual surfaces of the tooth roots. Internally, bone changes may manifest as either a reduction or increase in bone structure, with a reduction appearing as increased radiolucency (due to decreased trabeculae), and an increase showing as radiopacity (sclerosis), indicating thicker and denser trabeculae. Typically, both bone loss and formation occur in periodontal disease. Early acute lesions mainly exhibit bone loss, while chronic lesions show bone sclerosis. Various patterns of bone loss due to periodontitis can be observed in diagnostic images [1].

2.6. Computer Vision and Deep Learning

2.6.1. Overview

Computer Vision is a field of artificial intelligence (AI) that enables computers to interpret and understand visual information from the world, like human vision. It involves techniques such as image processing, object detection, facial recognition, and scene understanding. Computer vision applications can analyze images and videos for tasks like medical diagnostics, autonomous vehicles, surveillance, and augmented reality.

Supervised machine learning (ML) is a probabilistic learning method that uses training data to train a model that can be used for image segmentation and classification problems that are difficult to accomplish using classical image processing approaches. Examples of such tasks include distinguishing between images of cats and images of dogs or identifying handwritten digits.

Deep learning is a specialized subset of ML that focuses on using artificial neural networks to model complex patterns and relationships in data. While ML encompasses a broad range of algorithms (such as decision trees, support vector machines, and linear regression), deep learning specifically relies on multi-layered neural networks, often called deep neural networks.

Deep learning is a more advanced form of ML, particularly useful for tasks like image recognition, speech processing, and natural language understanding [10].

2.6.2. Computer Vision

The field of computer vision has many applications, a few of which are discussed here:

2.6.2.1. Image Classification

Image classification is the process of assigning a single label to an image based on its content. A DL model learns from labeled images to recognize patterns and classify new images accordingly.

Example: Identifying whether an image contains a cat, dog, or bird.

2.6.2.2. Object Detection

Object detection goes beyond classification by identifying multiple objects within an image and their locations. It draws bounding boxes around detected objects and labels them.

Example: Detecting pedestrians, traffic signs, and other vehicles.

2.6.2.1. Image Segmentation

In semantic image segmentation, the goal is to assign each pixel to a certain class, following which spatial clusters of similar pixels can be classified as objects. Figure 2-11 shows a segmented image, where the laptop is black, the charging cable is red, the cup is blue, the desk is yellow, and the background is white. This approach groups similar objects into a single category but doesn't distinguish between individual instances.

Example: Detecting all "cars" in an image but not differentiating between them.

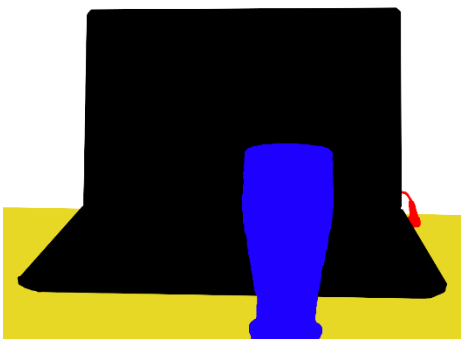


Figure 2-11 - Image segmentation

Adapted from https://commons.wikimedia.org/wiki/File:Image-segmentation-example-segmented.png

2.6.2.2. Instance Segmentation

Instance segmentation is like semantic segmentation but differentiates between individual instances of the same object. For example, multiple cars in a scene would be labeled as separate objects.

Recently, tools that enable relatively simple implementation of deep learning networks have become available. These include Python libraries such as TensorFlow [11] and Pytorch [12], applications such as MATLAB [13] and mianalyzer [14] and frameworks such as napari [15] and ImageJ [16] in which deep learning application can run. In parallel with the availability of easily used software tools, there are also fast GPU servers available that enable rapid training of deep learning networks, thereby greatly reducing the time required for development (especially training). We used these tools to develop a deep learning-based workflow for analysis of dental X-ray images.

2.6.3. Neural Networks and Convolutional Neural Networks

Deep learning is a subset of ML. It is inspired by the architecture of the neuron: the neuron receives a signal (input) from its dendrites, the dendrites send the signal to the cell body where it is processed (activation function). Then the signal travels along the axon to other dendrites or the target cells (output). The network of connections between dendrites and axons of the many neurons in the brain is called a neural network. Artificial neural networks inspired by biological neural networks are the basis of various deep-learning architectures. Deep learning systems do not require predefined feature sets.

ML can be either supervised, where the system learns from a data set with predefined ground truth, or unsupervised, where the system uses clustering algorithms without predefined ground truth. In this research, a supervised approach is used, and annotated images will be used to train the system.

The most promising algorithms for computer vision are based on convolutional neural networks (CNN) as shown by the AlexNet architecture [17]: this approach uses the convolution operator to implement multiple filters, and with the concatenation of various filters across various layers, the algorithm can learn the parameters which can best solve the problem.

2.6.3.1. Hyperparameters

Hyperparameters are a set of parameters which are used for regulating the design of the model (such as learning rate and regularization), they are different from the more fundamental parameters representing the weights of connections in the neural network [10].

- **Learning rate:** The learning rate determines how sensitive the model will be for correcting errors. A high learning rate can result in overshooting which may cause the model to not converge, while a small learning rate can make the model too slow to reach the minimal error.
- **Epochs:** The number of epochs is the total number of iterations are run to train the model. The model parameters (often tens of millions of them) are updated each epoch. A larger number of epochs will give the model more time to adapt but may result in over-fitting to the training dataset.
- **Batch size:** The batch size determines the number of images fed to the network **before** updating the weights of the model. A large batch size speeds training by processing more images in parallel but may result in out-of-memory errors in the GPU.

2.6.4. Architecture and Image Segmentation Model - U-net

U-Net [18] is a fully convolutional neural network primarily designed for semantic segmentation, especially in the field of medical image analysis. The model's architecture has a distinctive U-shape, hence the name "U-Net." It's designed to predict pixel-wise labels and is particularly known for its skip connections, which help preserve important spatial details.

U-net consists of the following components:

- **Encoder (Contracting Path):** The encoder part of U-Net works by progressively down sampling the input image to capture high-level features. It consists of several convolutional layers followed by max-pooling layers. Each convolution layer uses a small 3x3 kernel to extract features like edges, textures, and patterns. As the image is down sampled, the spatial resolution decreases, but the network captures more abstract features.

- **Decoder (Expanding Path):** The decoder works by progressively up sampling the feature maps to restore the original image's spatial resolution. The goal is to generate a pixel-wise classification of the input image. Each layer in the decoder consists of up-convolutions (also known as transpose convolutions or deconvolutions) to up sample the feature maps.
- **Skip Connections:** Skip connections pass feature maps from the encoder directly to the corresponding decoder layer. Skip connections allow the model to retain fine-grained spatial information that might be lost during down sampling. This helps with precise boundary delineation in the segmentation task, which is crucial for tasks like medical image segmentation (e.g., detecting tumors or organs).

2.7. Previous Work

The use of AI for computer vision tasks in the dental field has a rich history behind it.

One study attempted to predict posterior initial proximal caries in bitewing images [4] using CNN, as well as whether the proposed model can help clinicians diagnose dental caries in actual clinical settings. In this study, two experienced dentists annotated 304 bitewing radiographs by drawing the contours of each annotation. Two U-net CNN models were selected and trained using augmented data, one for caries detection, and another for structural detection. The study showed that there was a significant improvement in the dentist's ability to diagnose caries when using the deep learning model.

Another study [19] pursued the same goals as the one mentioned above, using two different networks: VGG-16 for object detection, and a U-net architecture for segmentation. Their data contained 613 anonymized bitewing radiographs with no artefacts, which were annotated by two dentists. The researchers compared the AI model to 5 dentists with various experience level, and even though the network did not perform better than the experts, the model did outperform in experience dentists,

One study [20] assessed the performance difference between an AI model (U-net) to dentists regarding detecting caries, by comparing the IOU metric of both sides. The dataset consisted of 3686 bitewing images. The study found that for detection of caries in bitewing radiographs, a deep neural network was significantly more accurate than dentists.

A different study [21] also used a U-Net base CNN for detecting dental pathologies. Using 500 bitewing images, the researchers claimed that the study has shown that an AI model can

be used to automatically evaluate bite-wing radiographs and the results are promising. Owing to these automatically prepared charts, physicians in a clinically intense environment will be able to work more efficiently and quickly.

The goals of this thesis are to add more data (more than 5,000 images), and to increase the number of classes (including important pathologies such as bone loss and radiolucent lesions [RL]) that are segmented (ten object classes will be labeled).

3. Materials and Methods

3.1. Ethical Approval and Bitewing Database

For this method, a large database of labeled images is needed. Therefore, this project is a retrospective study of images acquired by the IDF, in association with dentists who annotated the bitewing images.

This study was approved by the Institutional Review Board of IDF's Medical Corps, and all research was carried out in accordance with relevant guidelines and regulations.

3.2. Image Annotation

With the availability of easy-to-use deep learning systems, the major limitation in their application is the availability of large, annotated datasets. Since public domain annotated dental images are not available, we annotated images taken from the IDF, using the help of dentists. 6 dentists agreed to annotate the images, including 2 experts. An annotated dental image dataset would by itself be a significant contribution to the field. All annotations were done using Labelme software [22], and a total of 5,668 images were annotated.

The number of annotations for each class across all the images is:

1. Caries – 1,038
2. Tooth filling – 11,139
3. Secondary caries – 752
4. Bone loss – 6,006
5. Crown – 2,468
6. Impacted tooth - 476
7. Lesion RO – 102
8. Lesion RL – 613
9. Lesion Mixed – 13
10. Implant – 1,074

3.3. Hardware

The computer used to train the model had the following specifications:

Processor: 1x AMD Ryzen™ 7 5800H Processor (Ryzen™ 7 5800H)

Memory: 64GB RAM

Operating System: Windows 11 Home 64

Graphics Processing Unit (GPU): 1x NVIDIA® GeForce® RTX™ 3080 16GB RAM

3.4. Annotations Guidelines and Calibration

Since the interpretation of dental X-ray images can vary between different dentists, guidelines for annotating the different images were written by the most experienced dentist, with agreement of all the participating dentists.

A summary of the guidelines is presented here:

- **Annotation Categories**

Ten categories (classes) were annotated:

1. Primary caries (initial tooth decay)
2. Secondary caries (decay next to fillings)
3. Tooth fillings
4. Bone loss
5. Crowns
6. Dental implants
7. Impacted teeth
8. Radiolucent lesions (dark areas)
9. Radio-opaque lesions (white areas)
10. Mixed lesions (with both radiolucent and radio-opaque features)

- **General Annotation Principles**
  - Mark any finding in the image that requires dental treatment, even if only partly visible.

- Duplicate, erroneous, or in-treatment images should be moved to a “Not Relevant” folder.

- **What Not to Annotate**
  - Findings that are unclear or barely visible and cannot be diagnosed from the image alone
  - Missing teeth or teeth scheduled for extraction
  - Posts, fixation devices, or adhesive materials
  - Images with double exposures
  - Images taken during a procedure (e.g., length measurement, root canal in progress)
  - Non-diagnostic images or when a pathology cannot be identified with certainty
- **Caries (Tooth Decay)**
  - Only annotate decay that has penetrated the dentin layer; do not annotate initial enamel caries.
  - Cervical burnout (radiographic artifact) should not be marked unless actual caries is certain.
  - Secondary caries is decay near a filling; primary is decay away from a filling.
  - If it is unclear whether caries are primary or secondary, classify them as primary caries.

- **Tooth Fillings**
  - Includes all restorative materials such as amalgam and cast structures, but not bonding cement.
  - Multiple materials in the same tooth are annotated as a single filling region.
- **Bone Loss**
  - Mark areas where bone is missing and should normally be present.

    - Loss due to non-pathological causes (e.g., surgery) should be annotated.
    - Around implants, mark bone loss only from the first thread and downward.
- **Crowns**
    - Annotate only the metallic part of the crown, not porcelain or posts.
    - Connected crowns are annotated as one unit.
- **Impacted Teeth**
    - Annotate all visible parts of the impacted tooth.
    - If uncertain whether the object is an impacted tooth or a radio-opaque lesion, annotate as a lesion.
- **Lesions**
    - Classify as radiolucent (dark), radio-opaque (white), or mixed, as seen on the image.
- **Implants**
    - Annotate the implant body, even if not fully visible; highlighting the bone level around the implant is important.
    - Both precise “zig-zag” annotation and simple outline tracing are acceptable.

In addition to the guidelines (which were updated after each iteration of annotations from the dentists), a calibration was performed to ensure that all the dentists annotated the images uniformly. In the calibration sessions, all dentists were given the same 20 images chosen by an expert, each containing different pathologies to annotate. Then, the dentists would meet and discuss the difference in annotations, and each dentist would explain why he annotated the way he did. This was used to enforce uniformity of annotation, in order to reduce the difference between them as much as possible.

3.5. Software

In this research, Python version 3.9 for Windows 11 laptop was used, alongside Anaconda version 25.1.1, TensorFlow GPU version 2.10.1, Albumentations [23] version 1.2.0, and Segmentation Models version 1.0.1.

The annotation platform was Labelme [24]. Since Labelme is free and has a low learning curve, it was ideal for rapid annotations, without the need for high technical knowledge. The output of the labeling was a JSON file containing both the original image and the annotations.

3.6. Loss Functions & Evaluation Metrics

3.6.1. Focal Loss

Focal Loss is a loss function introduced for object detection tasks [24]. It was designed to address the class imbalance problem that occurs when training on datasets where the number of background (easy negative) pixels is much greater than foreground (positive) pixels. The formula for Focal Loss is given in eq. (1).

$$FL_{p_t} = -\alpha(1 - p_t)^{\gamma} * \log(p_t) \tag{1}$$

Where p_t is the model's estimated probability for the ground truth class, γ is the focusing parameter and α is a balancing weighting factor.

When an example is easy to classify (high p_t), the modulating factor becomes small, reducing its contribution to the total loss.

When an example is hard to classify (low p_t), the factor remains large, making the loss emphasize these samples.

3.6.2. Jaccard Index / Intersection Over Union

One of the metrics that was used to evaluate the network's performance is the Jaccard index / Intersection Over Union (IOU) [20]. Figure 3-1 shows overlap between two labels A and B. Suppose that A is the ground truth and B is the model's output. The IOU is calculated as:

$$IOU = \frac{Intersection}{Union} = \frac{A \cap B}{A + B - (A \cap B)} \tag{2}$$

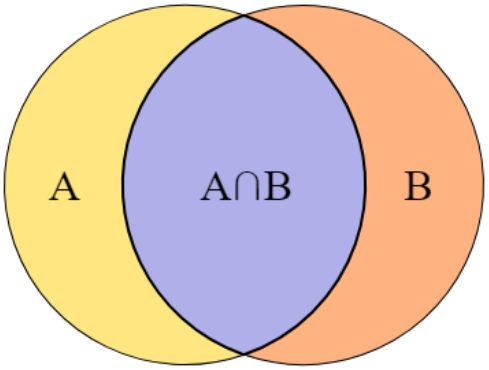


Figure 3-1 - Intersection Over Union

3.6.3. Dice Score / F1 Score

The Dice score / f1 score [19] measures the ratio between two times the intersection, divided by the total area of both labels. For the example in Figure 3-1, f1 is calculated as:

$$f1 = \frac{2 * (A \cap B)}{A + B} \tag{3}$$

The F1 score helps with class imbalance in image segmentation because it combines precision and recall (both are defined in the next section), instead of being dominated by the majority background pixels like accuracy. In highly imbalanced segmentation problems, a model that predicts almost all pixels as background can still achieve high accuracy, but it will have low precision or recall for the minority structures; the F1 score penalizes this by dropping sharply when either precision or recall is poor, making it more sensitive to the performance on small or rare objects.

3.6.4. Accuracy, Precision and Recall

The next three metrics [19] are computed per class (treating each pixel as a sample) from the confusion matrix counts: true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) [25].

Accuracy measures the proportion of all pixels (both positive and negative) that are correctly classified for that class or overall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

Precision measures how many of the pixels predicted as belonging to the class are correct (focuses on prediction quality among positives).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

Recall measures how many of the ground-truth class pixels were successfully recovered by the prediction (focuses on completeness of positives).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

In this research, semantic (pixel-wise) segmentation was used, and for every pixel a class label was assigned, to label the area where a pathology is located. In segmentation, IOU is computed between prediction and ground-truth masks, so slight boundary shifts, holes, or under-segmentation of thin structures can dramatically lower the IOU for a given class, even if the model has clearly identified that an object of that class is present in roughly the correct location. However, even though the dentists need to know the location of a pathology, the match does not have to be exact. Therefore, we also explored the detection rate of the objects in the model compared to the ground truth. In this manner, we can determine if the model can detect the objects, and to determine how much the model can help the dentists locate and identify pathologies. Here, we checked to see whether the centroid of a detected object fell inside the bounding box (BB) of a corresponding pathology in the ground truth. If yes, this would define a true positive (TP). If not, it would be a false positive (FP). The test was also run in the other direction (i.e. to test if the centroids of the ground truth objects fell within the bounding boxes of corresponding objects in the prediction image. Centroids in the ground truth that did not fall within a corresponding BB in the prediction image were false negatives (FN), In this project, by definition, there were no true negatives (TN), because TN pixels were by definition background. These metrics were used primarily for evaluating object detection, rather than for evaluation of the semantic segmentation.

3.7. Selected Models and Their Parameters

A segmentation model can be based on many architectures. We chose to use a U-net based architecture [18], since this architecture is considered well suited to semantic segmentation of images.

The library "Segmentation models" for python was used, within a TensorFlow framework. The segmentation models package includes a rich variety of sophisticated encoders for use with the U-net architecture. We experimented with different backbones/encoders such as

Resnet, Mobilenetv2 and Vgg19 and the backbone that produced the best results was inceptionv3 [26].

Transfer learning was used for the model, and the model was loaded with an inceptionv3 backbone pretrained on ImageNet.
The Adam optimizer was chosen with the parameters recommended by the author: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-7}$.
The models were trained for 200 epochs, with an Early Stop callback that halted training if overfitting was detected, when there was no improvement in the validation loss for 25 epochs. The learning rate for both models was 10^{-4} since this value produced the best results, and a learning rate reduce by a fator of 0.8 was used whenever the model did not improve after 5 epochs.

Data augmentation was provided by the Albumentations library [23]. The augmentations used in this project were:

- Vertical and horizontal flips (50% chance).
- Random brightness and contrast changes (70% chance).
- Rotation between -25° to +25° (70% chance).

Two metrics were used to evaluate the model: IOU and F1 Score. For both metrics, the Background class was omitted, while the other classes were equally weighted.

The model was trained using either Dice loss or Focal loss. For Focal loss, the parameters were $\alpha = 0.25, \gamma = 2$, as the authors of the paper recommend [27].

4. Results

4.1. U-net with Dice loss

The training set Dice loss was 0.25, IOU score was 0.66, and F1 Score was 0.73.

The validation set Dice loss was 0.34, IOU score was 0.55, and F1 Score was 0.62.

The graph of the loss function and the IOU metric is shown in Figure 4-1. As expected, with increasing epoch, the loss is decreased, while the IOU metric increased, both of which show that the model was learning.

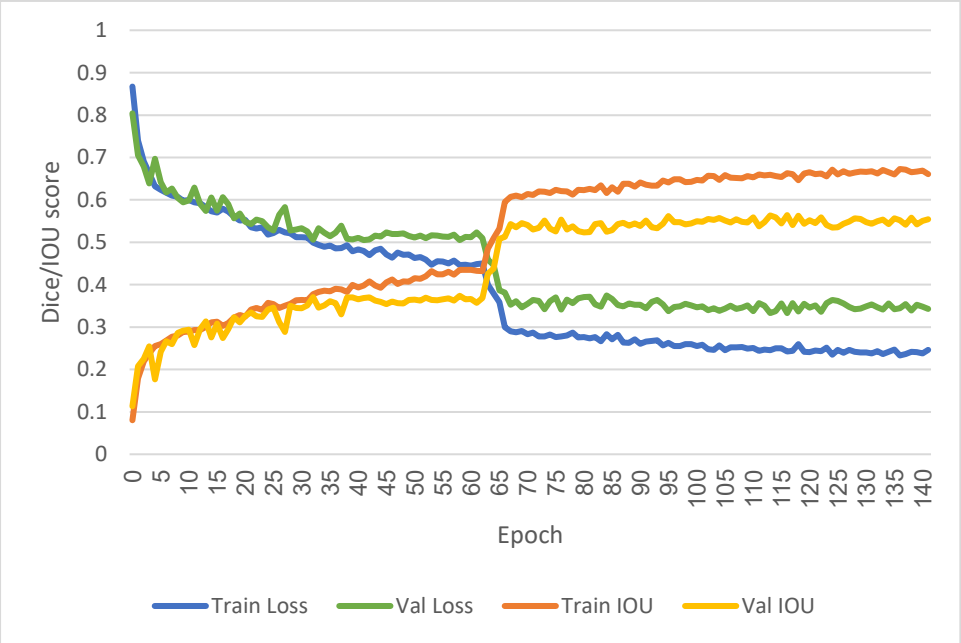


Figure 4-1 Loss and IOU for Dice loss

4.1.1. Metrics

The metrics for the training, validation and test sets are presented here. As expected, the metrics values for the training set are higher than the validation and test sets, and the metrics for the test set are lowest. Classes that have low contrast and small area such as caries tend to have lower scores compared to classes with high contrast and large area such as Tooth filling. Classes that had a low number of instances (like Lesion Mixed) have lower scores compared to classes with a large number of instances (Such as Crown). Tables 1-3 show the results for semantic segmentation of the images. In semantic segmentation, the IOU and F1/Dice metrics are more relevant compared to metrics such as precision, recall and accuracy, which are not good indicators at the pixel level. As a rule of thumb, for semantic segmentation, IOU of 50% or higher indicates a visually successful segmentation.

Table 1 - Training set Dice Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.985	0.658	0.831	0.590	0.612	0.885	0.783	0	0.638	0	0.828
f1 / Dice	0.993	0.794	0.908	0.742	0.759	0.939	0.879	0	0.779	0	0.906

Table 2 - Validation set Dice Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.980	0.357	0.798	0.282	0.390	0.856	0.601	0	0.392	0	0.751
f1 / Dice	0.990	0.526	0.888	0.440	0.561	0.923	0.751	0	0.563	0	0.858

Table 3 - Test set Dice Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.980	0.398	0.821	0.255	0.388	0.852	0.548	0	0.315	0	0.767
f1 / Dice	0.990	0.569	0.902	0.407	0.559	0.920	0.708	0	0.479	0	0.868

4.1.2. Typical Segmentation Results from the Test Set

Figures 4-2 through 4-7 show segmentation of test set images, compared to their ground truth. In these figures, the left image is the ground truth, and the right image is the prediction mask.

In Figure 4-2, the original image contains an RO lesion, 3 RL lesions, 3 tooth filings and 2 instances of bone loss. The image includes high contrast pathologies such as tooth fillings and Lesion RO, and lower contrast pathologies such as bone loss and Lesion RL. The model captured most of the annotations, with a few exceptions such as the Lesion RO, one Lesion RL and one bone loss. In addition, the model captured a tooth filling that the dentists did not annotate.

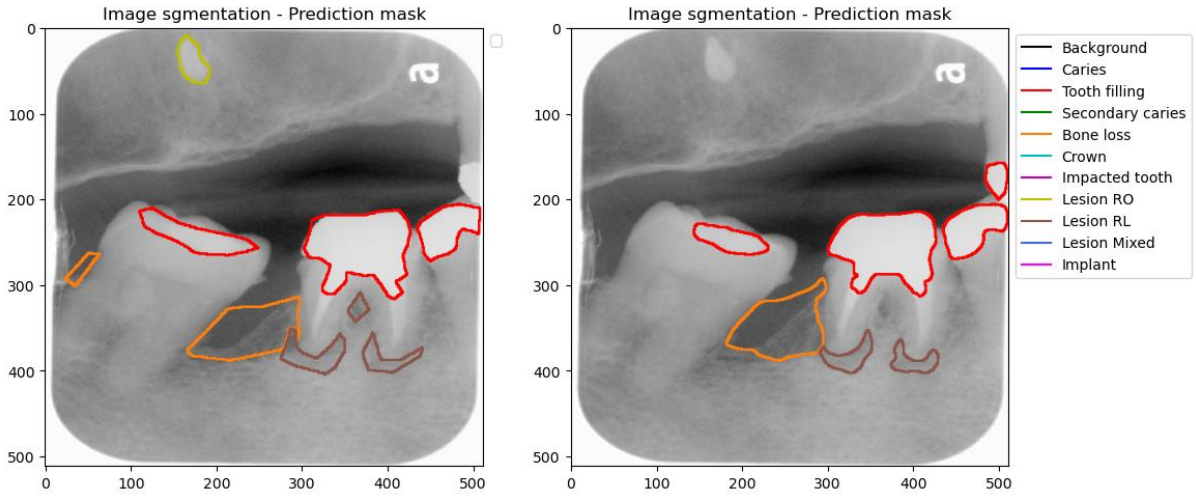


Figure 4-2 Image number 190 - Ground Truth vs Dice Loss

Figure 4-3 contains one tooth with a filling, and another tooth that has a crown and an implant. The model detected all 3 objects.

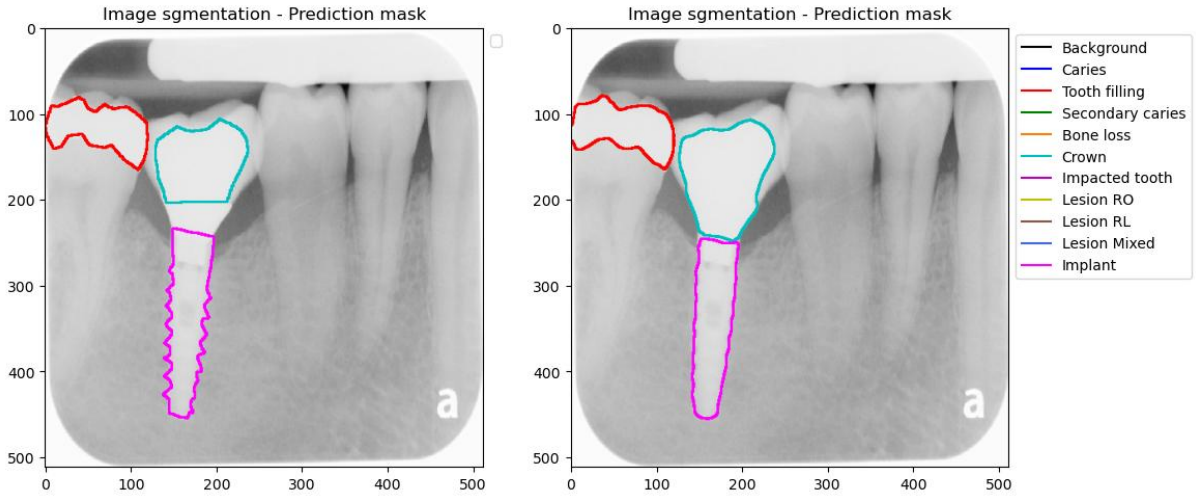


Figure 4-3 Image number 409 - Ground Truth vs Dice Loss

Figure 4-4 shows 3 teeth with a crown, 4 sites of secondary caries, 2 instances of bone loss and a filling. As expected, the model performed well segmenting the crown and fillings. The model also found the two instances of bone loss. The model captured three instances of secondary caries, but failed to capture the instance of secondary caries on the left side of the image, probably because it is small and located on the edge of the image. Instances of secondary caries, though detected, would not have a very high IOU score, since the pixel overlap of the detected object and the ground truth is not high. Nevertheless, this is a case of successful detection, because the object would be flagged for examination by a dentist.

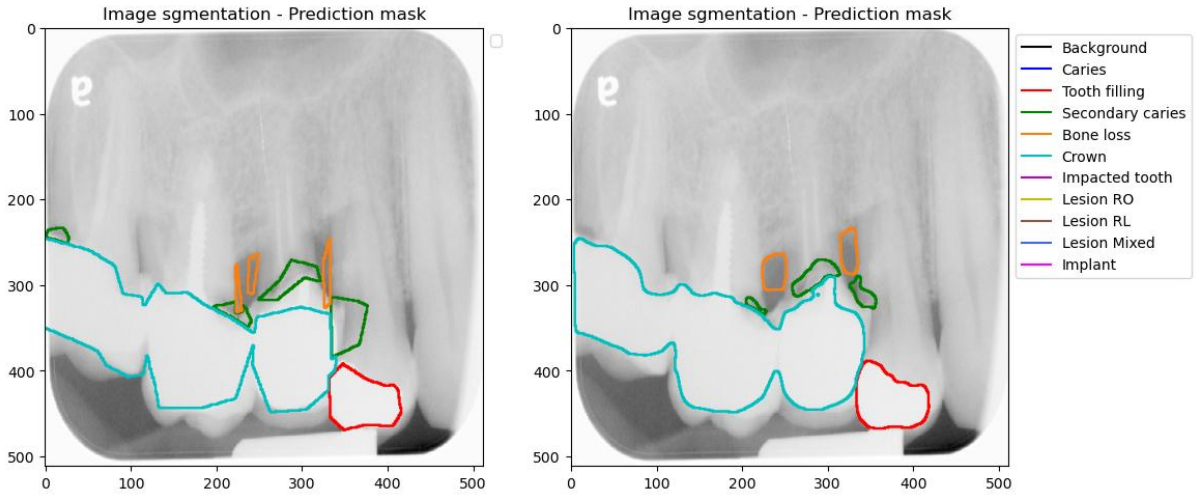


Figure 4-4 Image number 3,659 - Ground Truth vs Dice Loss

Figure 4-5 shows an image with caries in one tooth, as well as an RL lesion beneath the same tooth and in the tooth next to it. There are also 2 fillings and 3 instances of bone loss. The model captured the caries and the tooth fillings. The model captured 2 out of the 3 instances of bone loss. It captured a small portion of the large RL lesion but failed to detect the smaller RL lesion on the right, probably due to the low contrast.

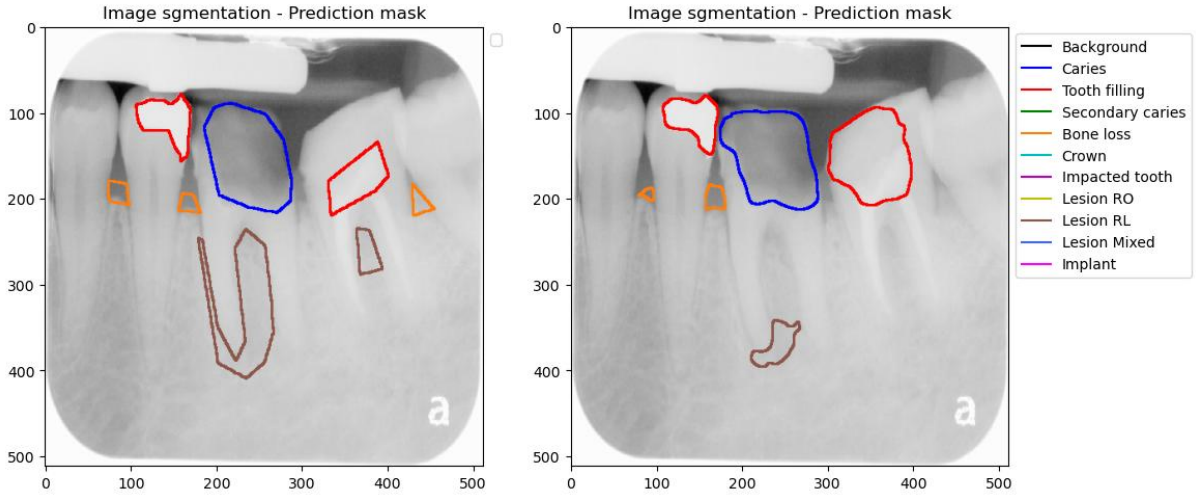


Figure 4-5 Image number 3,805 - Ground Truth vs Dice Loss

Figure 4-6 shows bone loss and an impacted tooth. Both are correctly segmented by the model.

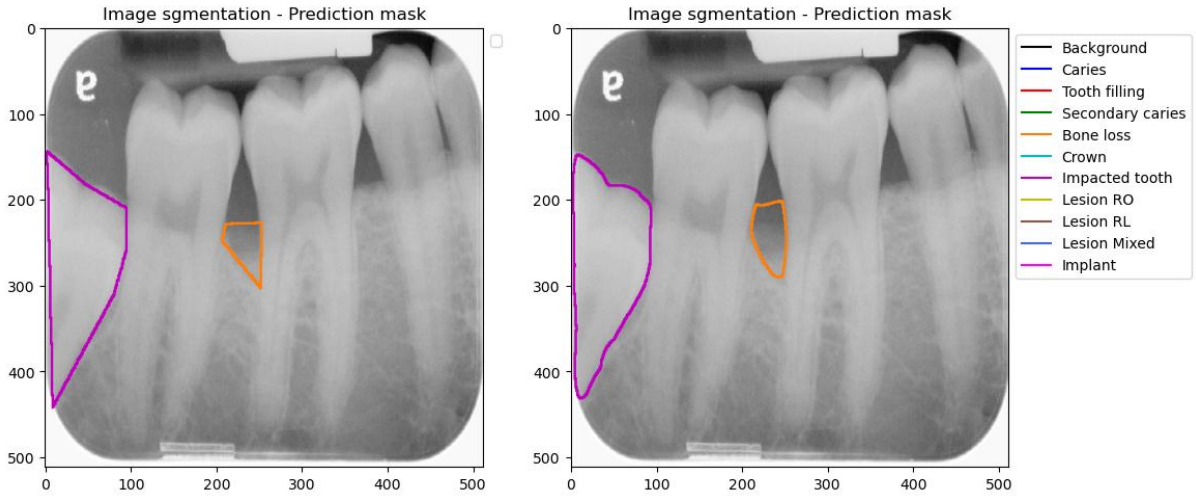


Figure 4-6 Image number 4,649 - Ground Truth vs Dice Loss

Figure 4-7 includes caries, tooth filing, an implant and a crown and a mixed lesion. The model captured the implant, crown and tooth filling with precision. However, the model failed to segment the mixed lesion and detected the caries in a different position altogether (false positive and false negative).

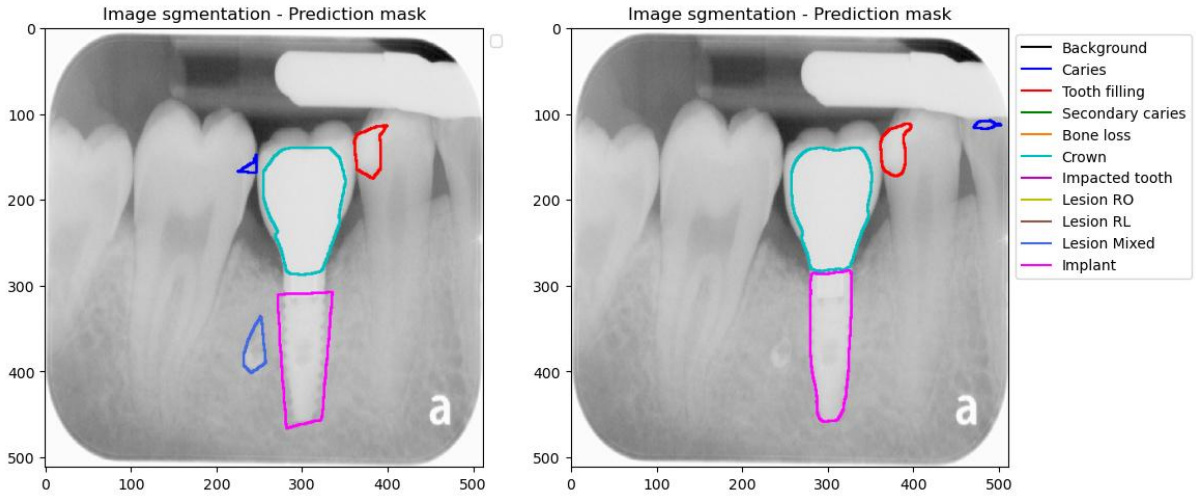


Figure 4-7 Image number 5,422- Ground Truth vs Dice Loss

4.2. U-Net with Focal loss

The training set Focal loss was $2.25 * 10^{-4}$, IOU score was 0.25, and F1 Score was 0.33.

The validation set Focal loss was $6.68 * 10^{-4}$, IOU score was 0.22, and F1 Score was 0.29.

The graph representing the loss function and the IOU metrics is shown in Figure 4-8.

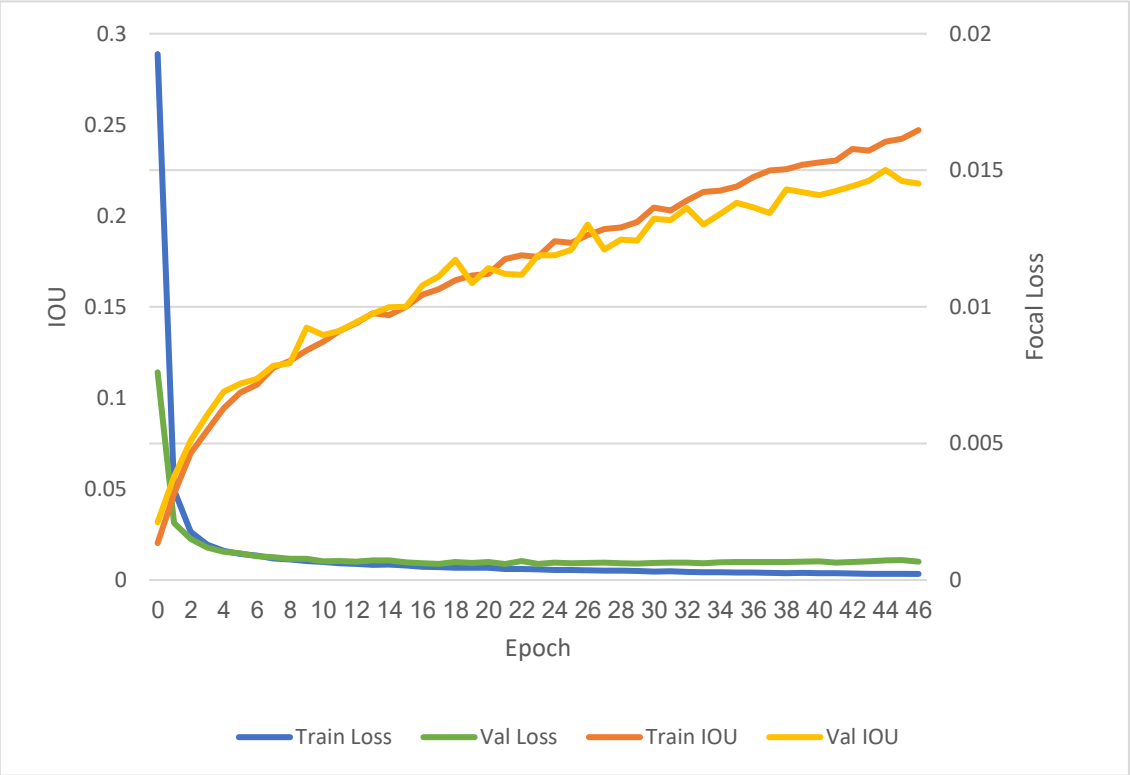


Figure 4-8 Loss and IOU for Focal loss

4.2.1. Metrics

The metrics for each of the training, validation and test sets are shown in Tables 4-6.

Table 4 - Training set Focal Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.988	0.586	0.873	0.529	0.583	0.922	0.834	0	0.590	0	0.870
f1 / Dice	0.994	0.739	0.932	0.692	0.737	0.959	0.909	0	0.742	0	0.930

Table 5 - Validation set Focal Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.981	0.339	0.811	0.240	0.386	0.842	0.561	0	0.288	0	0.747
f1 / Dice	0.990	0.506	0.896	0.388	0.557	0.914	0.719	0	0.447	0	0.855

Table 6 - Test set Focal Loss

Metric	Background	Caries	Tooth filling	Secondary caries	Bone loss	Crown	Impacted tooth	Lesion RO	Lesion RL	Lesion Mixed	Implant
IOU	0.981	0.394	0.823	0.176	0.374	0.850	0.490	0	0.361	0	0.770
f1 / Dice	0.990	0.566	0.903	0.299	0.545	0.919	0.658	0	0	0	0.870

4.2.2. Typical Segmentation Results from the Test Set

A few results from the test set are presented here (the same images that were used for the Dice Loss model), where the ground truth is shown on the left, the middle is the predicted mask from the Dice Loss model, and the prediction mask from the Focal Loss model is shown on the right.

Figure 4-9 shows an example where both the Dice Loss model and the Focal Loss model detected most of the classes that appear in the ground truth. However, the Focal loss predicted a smaller area for the bone loss on the left side, and it missed one of the secondary caries.

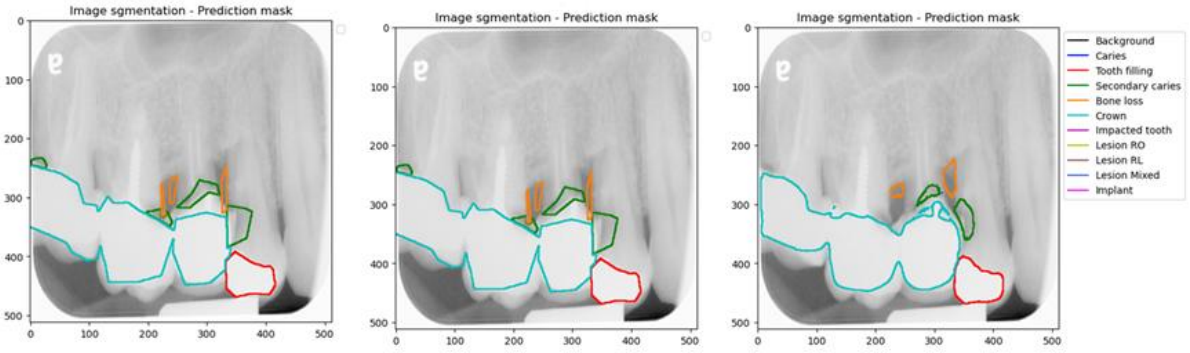


Figure 4-9 Image number 3,659 - Ground Truth vs Dice Loss vs Focal Loss

In Figure 4-10, the Dice Loss model and the Focal Loss model outputs are similar.

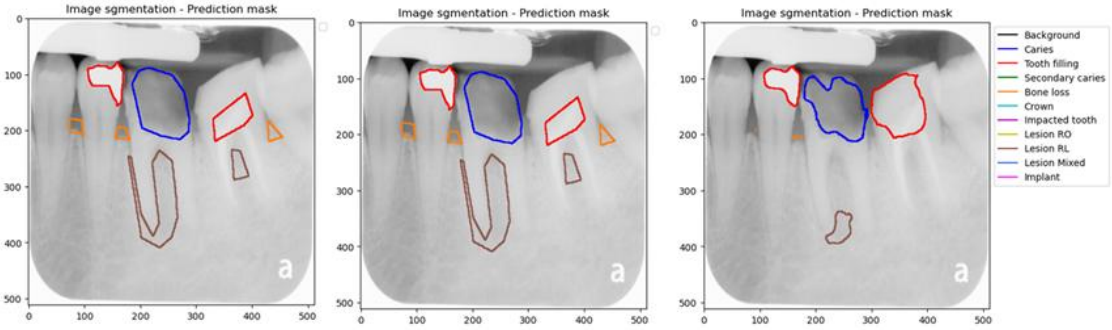


Figure 4-10 Image number 3,805 - Ground Truth vs Dice Loss vs Focal Loss

4.3. Matches Between Prediction and Ground Truth

The IOU metric calculates the overlap between different annotations. However, dentists don't require perfect overlap between objects, as long as the pathologies are detected in their correct locations. This is because once a pathology is flagged, a dentist must look at it in any case. Therefore, we tested whether the pathologies that were marked in the ground truth appeared in the predictions, even if the pixel level segmentation was not exact, as can be seen with the caries (blue annotation) in Figure 4-11. This was done using the predictions of the Dice loss model. The left image is the ground truth and the right image is the model's prediction.

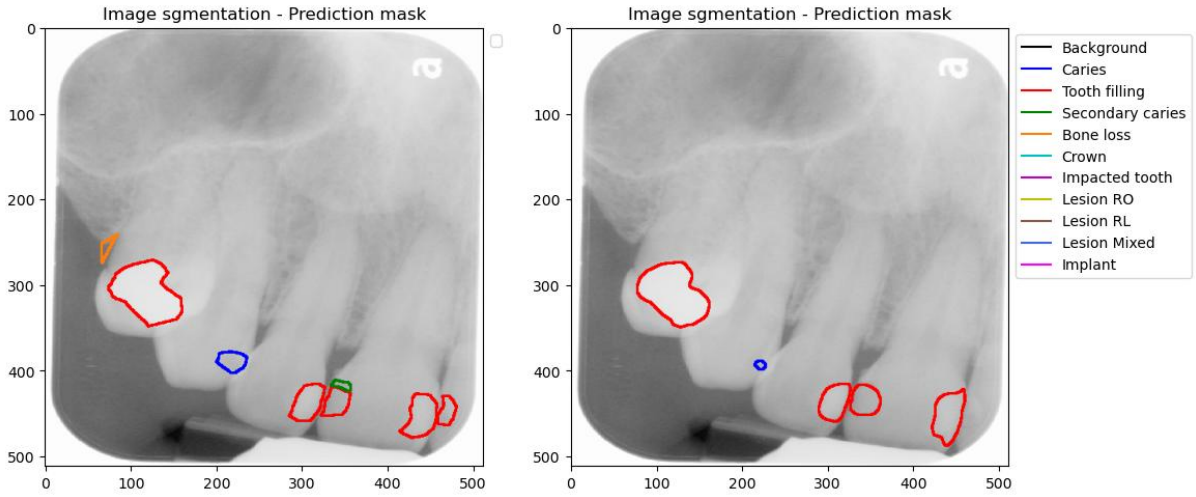


Figure 4-11 – Image 199: example of correct detection with inaccurate segmentation or the caries object.

An algorithm to determine the overlap of objects was used, as explained in the methods section. The results are presented in Table 7. All instances where overlap was found were considered as TP, the annotations that the model predicted which had no overlap were considered as FP, annotations from the ground truth that had no overlap in the model predictions were considered FN. Overlap was defined as having the centroid of an object in one image fall within the bounding box of a corresponding object in the corresponding image.

Table 7 – Objects detection, model's prediction and the ground truth

Metric / Class	TP	FP	FN	Precision	Recall	F1
Caries	67	57	75	0.54	0.47	0.50
Tooth filling	1,284	197	125	0.87	0.91	0.89
Secondary caries	32	66	85	0.33	0.27	0.30
Bone loss	492	372	371	0.57	0.57	0.57
Crown	336	35	18	0.91	0.95	0.93
Impacted tooth	55	32	16	0.63	0.77	0.70
Lesion RO	0	0	16	X	0	0
Lesion RL	37	35	54	0.51	0.41	0.45
Lesion Mixed	0	0	2	X	0	0
Implant	171	13	3	0.93	0.98	0.96

These results confirm that classes with many instances and high contrast such as tooth filings are detected more accurately than classes with a small number of instances and/or lower contrast such as primary caries (low contrast) or secondary caries (low contrast and low number of instances). Thus, filling, crown, and implant have high scores (over 0.87 for all metrics), while caries, secondary caries, bone loss, and RL lesions have scores in around 0.5. This is still much better than random guessing, meaning that the model has learned these structures to some extent. However, this is not good enough for clinical applications. Finally, secondary caries, lesion RO and lesion mixed do not have enough instances in the image set. For these classes, larger set of annotated images is required.

4.4. Valid Detected Objects Missed by the Dentists

Some instances were found where the model detected valid pathological objects which the dentists failed to annotate. One example is shown in Figure 4-12. This shows that in a real-world scenario, the model can support the dentists by capturing classes that they missed. This

finding is expected and is one of the motivations for using a DL model to screen dental image, especially in a scenario where many mages must be evaluated.

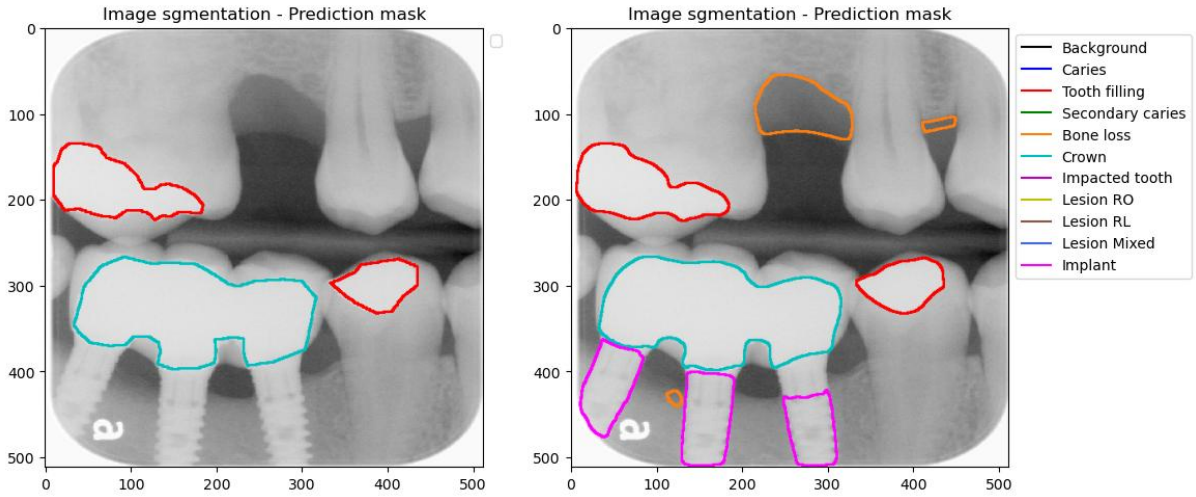


Figure 4-12 – Image 7,382, the Dice Loss model (right) captures pathologies missed by dentists (left)

5. Summary and Conclusions

5.1. Discussion

The goal of the thesis was to train a deep learning model to assist dentists with locating pathologies in dental x-ray images. In this research, more than 5,500 images were annotated with over 23,000 annotations.

Two models were trained using different loss functions (Dice loss and Focal loss). The loss curves (Figure 4-1 and Figure 4-8) show that both models learned to recognize dental pathologies. For both graphs, the loss function decreased, and the IOU metric increased, and the validation set for both graphs shows no indication of overfitting.

The IOU and F1 scores for the predictions of the Dice loss model on the test set (Table 3) indicate that the model performs well for high contrast classes, such as tooth filling, crown, and implant, where IOU and Dice/f1 are both between 0.767–0.920. Impacted tooth is also not bad, with IOU=0.548 and F1=0.708.
Caries and secondary caries are more challenging: their IOU values (0.398 and 0.255) and Dice/f1 scores (0.569 and 0.407) are lower than for fillings and crowns, suggesting more confusion with neighboring structures or small lesions. This is not surprising, because these are small, diffuse, low contrast objects.
Similarly, Bone loss and lesion RL have IOU values (0.388 and 0.315) and Dice/f1 scores (0.559 and 0.479), which is consistent with the difficulty of segmenting low contrast, diffuse objects.
Lesion RO, and lesion mixed did not have enough examples in the training set, and were effectively not detected, reflecting strong class imbalance and indicating that the model did not learn these classes.

For the model trained using Focal loss (Figure 4-8), the trend is similar to the results for Dice Loss, but the values are lower, which indicates that the model trained using Dice Loss performs better.

Visually assessment of the model predictions from the test set also shows that the model trained using Dice Loss performs better than the model trained using Focal Loss. Both have similar results, but the Dice Loss produces more accurate borders. However, we note that the model trained with focal loss might have done better with more epochs. In general, it is necessary to optimize hyperparameters such as learning rate and number of epochs.

The focus until now has been on semantic segmentation, but if a pathology is identified and located, it is not important that its boundaries and area be exact. An example of this can be seen in Figure 4-10 , where bone loss is detected, even though the area of the loss is not exact. This is an example where the IOU would be low, yet the detection is valid.

For this reason, we developed simple tools to compare the correlation between objects in the ground truth vs objects predicted by the model. We tested whether the centroid of predicted objects fell within the bounding box of a corresponding ground truth object. We measured precision, recall and F1 with respect to object detection. Table 7 shows the comparison of detections from the model predictions compared to the ground truth, and as expected, classes such as crown that cover a large area, have high contrast and have many instances, are more easily detected than classes such as secondary caries that cover a small area, have low contrast, and have fewer instances. An extreme example can be seen with the Lesion RO and Lesion Mixed classes, which the model does not detect. These two have few instances, and usually have a small area, and were not learned by the model.

A comparison of our results to previous studies appears in Table 8. Since our study includes more classes, our model performance is lower compared to the others.

Studies [4], [19] and [20] all were trained to detect a single class: caries. Each of them calculated the Precision, Recall and F1 Score metrics for detection of caries.

Table 8 - Comparision to different studies

Study \ Class	Precision	Recall	F1 Score
[4]	0.6329	0.6502	0.6414
[19]	0.84	0.84	0.84
[20]	Not given	0.75	0.73
Our study [Caries]	0.54	0.47	0.5

It may be that since our study had ten times the number of classes, the results for caries are lower compared to the other studies. A model trained only on caries might match the results of the other studies and may even surpass them since more images are available. Training separate models for difficult classes is one of the suggestions for further study that appears at the end of this section.

The study presented in [21] used 500 images, and trained on five classes, three of which overlap with the classes included in our study. Table 9 shows a comparison of the study in [21] to our study.

Table 9 - Comparison to study [21]

Class	[21]			Our study		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Caries	0.95	0.82	0.88	0.54	0.47	0.5
Crown	1	0.93	0.96	0.91	0.95	0.93
Restoration / filing	0.98	0.96	0.97	0.87	0.91	0.89

Both studies performed well for the crown and filling classes. The main difference is with the caries class, where our model's performance is poor compared to the other study. The number of total caries instances in the training set of [21] is 1,217. In our training set, even with

thousands of images, we had a similar number: 1,038.
The study in [21] selected for high quality images for training and testing, while our study incorporated a wide range of image quality. The data used by the study in reference 21 is not available, and therefore our model could not be directly compared to their model. We note that the study in [21] did not include classes such as bone loss, RL lesions, and secondary caries, which are all very important.

In some test set images, the model detected valid objects that the dentists themselves did not annotate. The agreement between dentists is subjective and even after 6 calibration procedures, misclassifications can still occur.

5.2. Future Work

This research presents a deep learning model that can detect dental pathologies. Ten classes were included in the study, and eight of them had enough instances to enable the model to predict them at greater than random chance. High contrast classes were detected well, while low contrast and diffuse classes were detected but not with sufficient accuracy and precision for clinical use. We offer the following suggestions for improvement of the model. These include:

1. Enlarge the dataset to include more instances of rare classes (e.g. radio opaque and radio-mixed lesions). More caries and secondary caries are also needed. It may be useful to use semi-automatic labeling tools such as QPath combined with SAM (segment anything model, developed by Meta).
2. Examine the effect of hyper parameters such as epochs, loss rate, and others on the performance of the model.
3. Examine other model architectures, backbones, and loss functions included in the Segmentation Models package. Loss functions optimized for cases of class imbalance should be tested.
4. It is possible that two separate models should be trained – one for high contrast easily identified classes, and one for diffuse low contrast classes such as bone loss and caries.
5. Examine whether instance segmentation using a package like YOLO, with annotations consisting of drawing bounding boxes around objects, would be more effective.

Our goal for the future is to enlarge the training set, and then use the approaches listed above to improve the model, so it can be used in the IDF as an aid to dental screening.

6. References

[1] R. C. S. a. G. Weiss, Woelfel's Dental Anatomy, Philadelphia: Lippincott Williams & Wilkins, 2012.

[2] A. Wenzel, "Bitewing and Digital Bitewing Radiography for Detection of Caries Lesions," *SAGE Publications,* 2016.

[3] M. N. ,. Y. H. ,. T. O. ,. M. B. D. ,. D. M. Anastasios Plessas, "Impact of time pressure on dentists' diagnostic performance," *Journal of Dentistry,* pp. 38-44, 2019.

[4] S. O. J. J. S. K. Y. S. &. J. P. Shinae Lee, "Deep learning for early dental caries detection in bitewing radiographs," *Scientific Reports,* 2021.

[5] A. I. I. N. B. P. Robert H Selwitz, "Dental caries," *The Lancet,* 2017.

[6] S. C. White and M. J. Pharoah, Oral Radiology Principles And Interpertation, Elsevier Inc., 2014.

[7] P. Erçal, "Prevalence of Impacted Teeth and Related Pathologies: A Retrospective Radiographic Study," *EssentDent.*

[8] D. Walmsley, Restorative Dentistry, Elsevier, 2007.

[9] S. E. Uribe, "Diagnostic agreement between radiologists, dentists and dental students for radiographic detection of approximal caries," *Journal of oral research,* 2014.

[10] C. C. Aggarwal, Neural Networks and Deep Learning, 2018.

[11] L. Yu, "Research and Implementation of CNN Based on TensorFlow," *IOP Conference Series: Materials Science and Engineering,* 2019.

[12] S. S. a. K. Sawant, "Face mask detection for covid_19 pandemic using pytorch in deep learning," *IOP Conference Series: Materials Science and Engineering,* 2021.

[13] S. Patil, "Algorithmic analysis for dental caries detection using an adaptive neural network architecture," *Heliyon,* 2019.

[14] N. Körber, "MIA: An Open Source Standalone Deep Learning Application for Microscopic Image Analysis," 2022. [Online]. Available: https://www.biorxiv.org/content/10.1101/2022.01.14.476308v1.

[15] n. contributors, "napari: a multi-dimensional image viewer for python," 2019. [Online]. Available: https://zenodo.org/record/3555620.

[16] C. A. R. W. S. &. E. K. W. Schneider, "NIH Image to ImageJ: 25 years of image analysis," *Nature Methods,* p. 671–675, 2012.

[17] A. Krizhevsky, "ImageNet Classification with Deep Convolutional Neural Networks," 2012.

[18] O. F. P. B. T. Ronneberger, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *Medical Image Computing and Computer-Assisted Intervention – MICCAI ,* 2015.

[19] I. S. Bayrakdar, "Deep-learning approach for caries detection and segmentation on dental bitewing radiographs," *Oral Radiology,* 2022.

[20] A. G. Cantu, "Detecting caries lesions of different radiographic extension on bitewings using deep learning".

[21] O. Baydar, "The U-Net Approaches to Evaluation of Dental Bite-Wing Radiographs: An Artificial Intelligence Study," *Diagnostics,* 2023.

[22] K. Wada, "Labelme: Image Polygonal Annotation with Python," 7 November 2018. [Online].

[23] A. P. E. K. V. I. a. A. K. A. Buslaev, "Albumentations: fast and flexible image augmentations," *ArXiv e-prints,* 2018.

[24] K. Wada, "labelme: Image Polygonal Annotation with Python," 2018. [Online]. Available: https://github.com/wkentaro/labelme.

[25] J. Jordan, "Evaluating image segmentation models," 30 05 2018. [Online]. Available: https://www.jeremyjordan.me/evaluating-image-segmentation-models/.

[26] C. Szegedy, "Rethinking the Inception Architecture for Computer Vision," 2015.

[27] T.-Y. e. a. Lin, " Focal loss for dense object detection," *Proceedings of the IEEE international conference on computer vision,* 2017.

[28] M. M.KASSAB, "The etiology and prevalence of gingival recession," *The Journal of the American Dental Association,* pp. 220-225, 2003.

[29] Y. W. a. A. K. a. F. M. a. W.-Y. L. a. R. Girshick, "Detectron2," 2019. [Online]. Available: https://github.com/facebookresearch/detectron2.

7. Appendix

7.1. Code

The full code and custom libraries are available on GitHub:

https://github.com/avieliyahu/Dental_AI_BIU_IDF

7.2. Dataset

The images and masks are available on Google Drive:

https://drive.google.com/drive/folders/1uSGqFGOKf5b-nYH8-R8GalEzyuoxu9Wq?usp=sharing

תקציר

מחקר זה הינו מחקר רטרוספקטיבי אשר מטרתו היא לאמן מודל למידה עמוקה אשר יוכל לזהות פתולוגיות וממצאים דנטליים בתמונות רנטגן דנטליות. במחקר, 6 רופאי שיניים תייגו מעל 5,500 תמונות נשך ותמונות פריאפיקליות על מנת לאמן את מודל הלמידה העמוקה. המודל הנבחר הינו מודל U-net אשר השתמש במשקולות מאומנות מראש (Transfer learning) ונעזר בCallbacks בכדי לעצור את המודל לפני שייגרם Overfitting, על מנת לטייב את המודל. בחינת תוצאות המודל מראות שהמודל אכן מסוגל לזהות פתולוגיות דנטליות בדיוק רב, ובמידה והמודל ייושם, הוא יוכל לפתוח צוהר לתיוג של תמונות נוספות ושיפור המודל.

<u>תודות</u>

המחקר הזה יכל היה להסתיים אי שם באיזור דצמבר 2023, אך לצערי, המתקפה הברברית של חמאס ב07/10/2023 קטעה את כל רצף העבודה. במהלך שנתיים של לחימה עזה, בה אני בתור קצין בדרגת סרן בצה״ל ושאר רופאי השיניים הצבאיים שלקחו חלק במלחמה עשינו ככל יכולתנו לתרום למאמץ המלחמתי, הצלחנו לסיים עם מחקר. אני מרכין ראש לזכר כל הנופלים במלחמה הנוראית הזו, ורוצה לשלוח תנחומים למשפחות הנופלים ורפואה שלמה לכלל הפצועים.
עכשיו לתודות:

בראש ובראשונה, ארצה להודות לבורא עולם שנתן בי את הבינה והכוח ללמוד ולחקור על מנת לבצע את העבודה הזו.

בנוסף, ארצה להודות להוריי היקרים גיל וטובה אליהו, שמאז שאני זוכר את עצמי הם עודדו אותי לרכוש השכלה ולא להפסיק ללמוד.

כמובן, ארצה להודות רבות לפרופסור אריה וייס, שליווה אותי במהלך כל המחקר, נתן עצות שלא יסולאו בפז, ולמד דברים חדשים במהלך המחקר ושיתף אותי בהם.

ארצה גם להודות לעמיתיי ומפקדיי בצבא ההגנה לישראל:
לתא״ל פרופסור אלון גלזבר, אל״מ ברק גורדון ולאע״צ ראיד כיוף על אישור הפרויקט בתוך חיל הרפואה.
למפקדים שלי רס״ן דור בוגנים הררי, סא״ל מיטל צור וסא״ל שרי ברדה על התמיכה הרבה בפרויקט ועידוד לביצועו תוך כדי תפקידי בצה״ל.
לסרן ישעיה ווידמן על כך שהוא הלך איתי לאורך כל הפרויקט, תמך בי ועזר רבות לפיתוח האלגוריתם.
לסא״ל ד״ר אבי שמש על האישור לביצוע המחקר בתמיכת רופאי השיניים של צה״ל ושימוש במאגרי המידע הצבאיים.
לרס״ן ד״ר אלכס לבובסקי על הובלה של תיוג התמונות ותמיכה מקצועית בכל התחום הדנטלי, ולכל שאר רופאי השיניים שעמלו ועבדו איתנו.

עבודה זו נעשתה בהדרכתו של פרופסור אריה וייס מהפקולטה להנדסה של אוניברסיטת בר-אילן.

אוניברסיטת בר-אילן

זיהוי פתולוגיות דנטליות באמצעות למידה עמוקה

אביאל אליהו

עבודה זו מוגשת כחלק מהדרישות לשם קבלת תואר מוסמך
בפקולטה להנדסה של אוניברסיטת בר אילן

רמת גן תשפ״ו

אוניברסיטת בר-אילן

זיהוי פתולוגיות דנטליות באמצעות למידה עמוקה

אביאל אליהו

עבודה זו מוגשת כחלק מהדרישות לשם קבלת תואר מוסמך
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רמת גן תשפ״ו